

Project Proposal

Spatiotemporal Graph Neural Networks for Real-Time Fault Localization in Smart Grids

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1 Introduction

The transition to “Smart Grids” is critical for integrating renewable energy sources and ensuring a sustainable future (Green AI). However, modern power grids are becoming increasingly complex due to the stochastic nature of renewable generation (e.g., solar and wind) and the bidirectional flow of power. In such dynamic environments, traditional fault detection methods—which often rely on manual rule-based thresholds or simple statistical models—are becoming insufficient. They frequently fail to capture the complex, non-linear spatial dependencies between grid nodes (buses) and the temporal dependencies of fluctuating sensor data.

This project proposes a deep learning approach using **Spatiotemporal Graph Neural Networks (ST-GNN)**. Unlike standard Convolutional Neural Networks (CNNs) that treat data as Euclidean grids (images), GNNs are designed to process data represented as graphs. Since a power grid is naturally a graph (where buses are nodes and transmission lines are edges), GNNs offer a theoretically superior method for modeling the system’s physical topology to detect and localize faults in real-time.

2 Problem Statement

Grid failures, if not detected and isolated within milliseconds, can lead to cascading blackouts, equipment damage, and significant economic loss. The core challenges are:

- Topological Complexity:** Faults at one node propagate effects across the network based on physical connections, which standard ML models ignore.
- Data High-Dimensionality:** Smart meters generate massive streams of time-series data that are difficult to process in real-time.

3. ****Localization Precision:**** Detecting *that* a fault occurred is easier than pinpointing *exactly where* it happened (e.g., Bus 14 vs. Bus 15), which is crucial for maintenance crews.

3 Literature Review

Recent research has shifted from traditional SVM and Random Forest models toward deep learning. Liao et al. (2021) highlighted that while CNNs are effective for time-series, they fail to model the irregular connections of power systems. Owerko et al. (2020) demonstrated the efficacy of GNNs for Optimal Power Flow, proving that graph-based models respect physical constraints better than fully connected networks. Furthermore, Chen et al. (2020) proposed GCNs for fault diagnosis but focused primarily on classification rather than precise localization. Our work builds upon the “Spatiotemporal” concept introduced by Yu et al. (2018), aiming to combine Graph Convolution (spatial) with LSTM (temporal) to address the localization gap in renewable-heavy grids.

4 Objectives

The primary goal of this research is to build an AI framework capable of identifying grid faults with high spatial precision.

- **Objective 1:** To model the IEEE standard bus systems (14-bus, 39-bus) as graph structures using adjacency matrices.
- **Objective 2:** To design and implement a Spatiotemporal Graph Neural Network (ST-GNN) architecture combining Graph Convolutional Networks (GCN) and Long Short-Term Memory (LSTM) units.
- **Objective 3:** To evaluate the model’s ability to classify fault types (e.g., Single Line-to-Ground) and localize the exact faulted bus index under varying noise conditions.

5 Proposed Methodology

The proposed framework consists of three stages:

5.1 1. Graph Construction & Preprocessing

We will represent the power grid as a graph $G = (V, E)$.

- **Nodes (V):** Represent buses/substations. Node features X_t will include voltage magnitude (V_m), phase angle (θ), and active power (P).
- **Edges (E):** Represent transmission lines. The connectivity is defined by the Adjacency Matrix A .

5.2 2. Model Architecture (ST-GCN)

The model will process data in two dimensions:

- **Spatial Block (GCN):** Aggregates information from neighboring nodes to capture topological dependencies.

$$H^{(l+1)} = \sigma(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} H^{(l)} W^{(l)})$$

- **Temporal Block (LSTM):** The output of the GCN is fed into an LSTM layer to capture time-series patterns (pre-fault vs. post-fault signal changes).
- **Output Layer:** A Softmax classifier that outputs the probability of a fault occurring at each node index.

5.3 3. Training and Validation

The model will be trained using Supervised Learning. We will use Cross-Entropy Loss to minimize classification error.

6 Dataset and Tools

- **Dataset:** We will use the **IEEE 14-bus and IEEE 39-bus test systems**. Since real fault data is scarce, we will generate a synthetic dataset using **SimBench** or **Pandapower** (Python libraries) by simulating various short-circuit faults (Single Line-to-Ground, Line-to-Line) at different locations.
- **Tools:**
 - **Language:** Python 3.9
 - **Deep Learning:** PyTorch Geometric (PyG) for GNN implementation.
 - **Simulation:** Pandapower for generating power flow data.

7 Expected Outcomes

We expect to develop a model that achieves:

- > 95% accuracy in distinguishing between normal operation and fault conditions.
- High precision in localizing the specific faulted bus, outperforming traditional CNN baselines.
- Robustness against measurement noise, simulating real-world sensor inaccuracies.

8 References

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